



SHORE

Exam Number:

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Year 12

Mathematics Extension 2

HSC Trial Examination

2022

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board approved calculators may be used
- Answer Questions 1 - 10 on the Multiple Choice answer sheet provided
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations
- Start each of Questions 11 – 16 in a new writing booklet
- Write your examination number on the front cover of each booklet to be handed in
- A NESA reference sheet is provided.
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”

Total marks – 100

Section I

Pages 2 - 5

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II

Pages 6 - 14

90 marks

- Attempt Questions 11 – 16
- Allow about 2 hours 45 minutes for this section.

Note: Any time you have remaining should be spent revising your answers.

**DO NOT REMOVE THIS PAPER FROM THE EXAMINATION
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Section I

10 marks

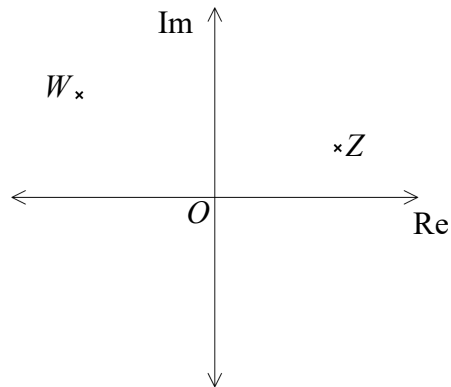
Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 Which expression is equal to $\int \frac{1}{x^2 + 6x + 13} dx$?
- (A) $\frac{1}{\sqrt{13}} \tan^{-1} \left(\frac{x+3}{\sqrt{13}} \right) + C$
- (B) $\frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + C$
- (C) $\frac{1}{4} \ln \left| \frac{x+1}{x+5} \right| + C$
- (D) $\frac{1}{4} \ln \left| \frac{x+5}{x+1} \right| + C$
- 2 What is the angle to the nearest degree subtended at the origin by the points (2, -1, 3) and (-4, 2, 2)?
- (A) 12°
- (B) 77°
- (C) 90°
- (D) 103°
- 3 The velocity of a particle is given by $v = 3 + x \text{ ms}^{-1}$. What is the acceleration, in ms^{-2} , of the particle at $x = 2 \text{ m}$?
- (A) 1
- (B) 3
- (C) 5
- (D) 8

- 4 The diagram shows the position of points Z and W , that have position vectors z and w respectively. In which quadrant would the point with position vector $\bar{z} - iw$ lie?



- (A) 1st
(B) 2nd
(C) 3rd
(D) 4th
- 5 What is the contrapositive of the statement:
'If Rashford plays and Man United win, I will be a happy chappy'.
- (A) If Rashford does not play or Man United do not win, I will not be a happy chappy
(B) If Rashford plays and Man United win, I will not be a happy chappy
(C) If I am a happy chappy, Rashford played and Man United won
(D) If I am not a happy chappy, Rashford did not play or Man United did not win

6 The vector equation of a sphere is given by $\left| \mathbf{z} - \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix} \right| = 52$. Where does the point with

position vector $\begin{pmatrix} -3 \\ 0 \\ 2 \end{pmatrix}$ lie with respect to the sphere?

- (A) At the centre of the sphere
- (B) Within the sphere, but not at the centre
- (C) On the surface of the sphere
- (D) Outside the sphere.

7 Which of the following statements is true?

- (A) $\forall a, b \in \mathbb{R} \quad \sin a = \sin b \Rightarrow a = b$
- (B) $\forall a, b \in \mathbb{R} \quad |a + b| > |a - b|$
- (C) $\exists a, b \in \mathbb{R} \quad \text{such that } \log_e(a + b) = \log_e(ab)$
- (D) $\exists a, b \in \mathbb{C} \quad |a + b| > |a| + |b|$

8 Given that $z = 2 - 3i$ is a root of $z^3 - pz^2 + 5z + q = 0$ where p and q are real, what are the values of p and q ?

- (A) $p = -2, q = -26$
- (B) $p = -2, q = 26$
- (C) $p = 2, q = -26$
- (D) $p = 2, q = 26$

- 9 It is given that $f(x)$ is continuous in the interval $-a \leq x \leq a$, and that $\int_{-a}^a f(x) dx = 0$.

Which of these does NOT follow?

(A) $\lim_{\varepsilon \rightarrow 0} \left(\int_{-a}^{\varepsilon} \frac{dx}{f(x)} + \int_{\varepsilon}^a \frac{dx}{f(x)} \right) = 0$

(B) $\int_{-a/2}^{a/2} f(2t) dt = 0$

(C) $\int_{c-a}^{a+c} f(x-c) dx = 0$

(D) $\int_{-a}^a f(x) + c dx - 2ac = 0$

- 10 Consider the line segments given by the parametric vector equations below.

$$\vec{r} = \begin{pmatrix} -3 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 2 \end{pmatrix}, \quad -2 \leq \lambda \leq 2;$$

$$\vec{l} = \begin{pmatrix} -2 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 4 \\ 6 \end{pmatrix}, \quad 0 \leq \mu \leq 2;$$

$$\vec{v} = \gamma \begin{pmatrix} 9 \\ -6 \end{pmatrix}, \quad 1 \leq \gamma \leq 3$$

Which of the following statements about the line segments is NOT true?

- (A) At least two of them are parallel
- (B) At least two of them are perpendicular
- (C) At least two of them have the same intercept
- (D) At least two of them have the same length

End of Section I

Section II

90 marks

Attempt Questions 11 - 16

Allow about 2 hours 45 minutes for this section

In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Answer question in a SEPARATE writing booklet.

- (a) $z = 2 + i$ and $w = 3 - 4i$. Evaluate $z\bar{w}$. 2
- (b) Given that $a, b \in \mathbb{N}$ and that a is a multiple of 6 and b is a multiple of 4, prove that $2a + 3b$ is a multiple of 12. 2
- (c) Evaluate $\int_0^{\frac{\pi}{4}} \tan^3 x \sec^2 x \, dx$. 2
- (d) Find $I = \int e^x \sin x \, dx$. 3
- (e) Sketch the curve described by $\frac{1}{z} + \frac{1}{\bar{z}} = 1$ on the complex plane. 3
- (f) Find the complex roots of $z^3 + 8i = 0$, and show them on an Argand diagram. 3

Question 12 (15 marks) Answer question in a SEPARATE writing booklet.

- (a) Show that the line given by parametric vector equation $\underline{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ 3

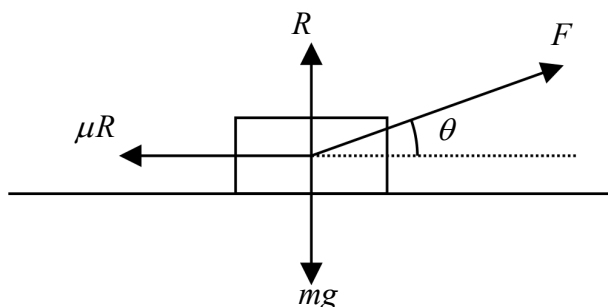
intersects the curve given by $\underline{r} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} t \\ t \\ t^2 \end{pmatrix}$ twice.

- (b) Prove by contradiction that for rational non-zero a and b and irrational x , that $ax + b$ is irrational. 3

- (c) Use the substitution $t = \tan \frac{\theta}{2}$ to find $\int \frac{1}{1 + \sin \theta + \cos \theta} d\theta$. 3

- (d) Prove that $x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$ for real $x \geq 0$. 3

- (e) An object with mass m is being pulled along horizontal ground with acceleration a by a constant force F acting at angle θ to the horizontal, as shown in the diagram. 3
A force due to gravity of mg acts downwards and a normal force R acts upwards and a resistive force μR acts parallel to the ground to oppose the motion, where μ , the coefficient of friction, is a constant.



By balancing horizontal and vertical components of forces, show that $\mu = \frac{F \cos \theta - ma}{mg - F \sin \theta}$.

Question 13 (15 marks) Answer question in a SEPARATE writing booklet.

- (a) Determine with working whether $(4, -1, 5)$ lies on the line with vector equation 1

$$\vec{r} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 2 \\ 5 \end{pmatrix}.$$

- (b) Evaluate: 2

$$1 + e^{\frac{2\pi i}{3}} + \left(e^{\frac{2\pi i}{3}}\right)^2 + \left(e^{\frac{2\pi i}{3}}\right)^3 + \dots + \left(e^{\frac{2\pi i}{3}}\right)^{3n}.$$

- (c) An object of mass 1 kg is projected vertically upwards from the origin with initial velocity V_0 , so that the only forces acting on it are a force due to gravity and air resistance, which is has size $\frac{\dot{y}}{10}$. Taking y as the displacement upwards from the origin, and acceleration due to gravity to be 10 ms^{-2} , its equation of motion is: 3

$$\ddot{y} = -\left(10 + \frac{\dot{y}}{10}\right)$$

Show that the maximum height the object reaches, $y_{\max}$ is

$$y_{\max} = 10 \left(V_0 - 100 \ln \left(\frac{100 + V_0}{100} \right) \right).$$

- (d) (i) Prove that $e^{in\theta} + e^{-in\theta} = 2 \cos n\theta$. 1

- (ii) Use part (i) and the binomial expansion of $(e^{i\theta} + e^{-i\theta})^3$ to prove that 2

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta.$$

- (iii) Hence find all solutions to $8x^3 - 6x - 1 = 0$. 3

- (e) Given that $u_{n+1} = 2u_n + 3$ and $u_1 = 1$, use mathematical induction to prove that $u_n = 2^{n+1} - 3$ for all positive integers n . 3

Question 14 (15 marks) Answer question in a SEPARATE writing booklet.

- (a) For all natural numbers n , if $(n-5)^2$ is an even integer, prove by contrapositive that n is odd. 2

- (b) Given that $\hat{a} = x\hat{i} + y\hat{j} + \frac{\sqrt{7}}{4}\hat{k}$, $\hat{b} = (x-2)\hat{i} + y\hat{j} + \frac{15}{4\sqrt{7}}\hat{k}$ and $\hat{a} \cdot \hat{b} = 0$, determine the value(s) of x and y . 3

- (c) Find $\int \frac{x}{1+\sqrt{x}} dx$. 3

- (d) A motorbike that is initially stationary at the origin has mass 400 kg and begins to travel horizontally with a constant propulsive force of 800N. A resistive force opposes the object's motion, proportional to the square of the velocity, such that its equation of motion is:

$$400\ddot{x} = 800 - kv^2$$

Where k is a constant, v is its velocity, and x is displacement in the direction of the motion after t seconds.

- (i) Show that $\frac{1}{a^2 - b^2} = \frac{1}{2a} \left(\frac{1}{a+b} + \frac{1}{a-b} \right)$. 1

- (ii) Given that the terminal velocity of the motorbike is 40 ms^{-1} , show that the object's velocity is given by 3

$$v = 40 \left(\frac{e^{0.1t} - 1}{e^{0.1t} + 1} \right).$$

- (iii) Find an expression for displacement x in terms of t . 3

Question 15 (15 marks) Answer question in a SEPARATE writing booklet.

(a) Let $I_n = \int_0^{\frac{\pi}{4}} \cos^n x \, dx$.

(i) Show that $I_n = \frac{1}{n} \left(\frac{1}{\sqrt{2}} \right)^n + \frac{n-1}{n} I_{n-2}$ for $n \geq 2$. **3**

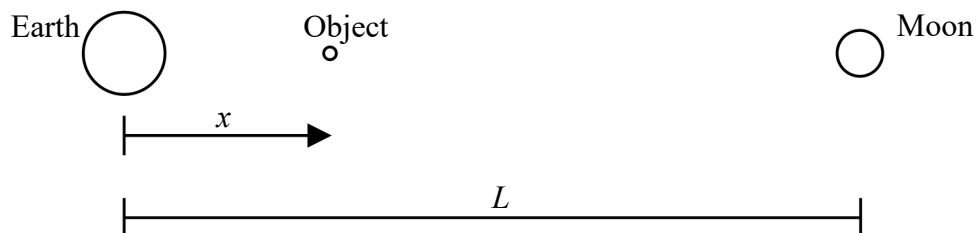
(ii) By using an appropriate trigonometric substitution, show that **2**

$$\int_0^2 \frac{1}{\sqrt{(x^2+4)^7}} \, dx = \frac{1}{64} \int_0^{\frac{\pi}{4}} \cos^5 \theta \, d\theta.$$

(iii) Hence deduce that $\int_0^2 \frac{1}{\sqrt{(x^2+4)^7}} \, dx = \frac{43}{3840} \left(\frac{1}{\sqrt{2}} \right)$. **2**

Question 15 continues on page 11

- (b) Consider an object travelling from the Earth to the moon. The distance between the centres of the Earth and moon is L and can be assumed to remain constant. The distance of the object from the centre of the Earth in the direction of the moon is x , where $0 < x < L$. Initially the object is at $x = 0.5L$ and is travelling with velocity $u \text{ ms}^{-1}$, where $u > 0$.



The equation of motion of the object is given by:

$$\ddot{x} = -\frac{81k}{x^2} + \frac{k}{(L-x)^2}, \text{ where } k \text{ is a constant.}$$

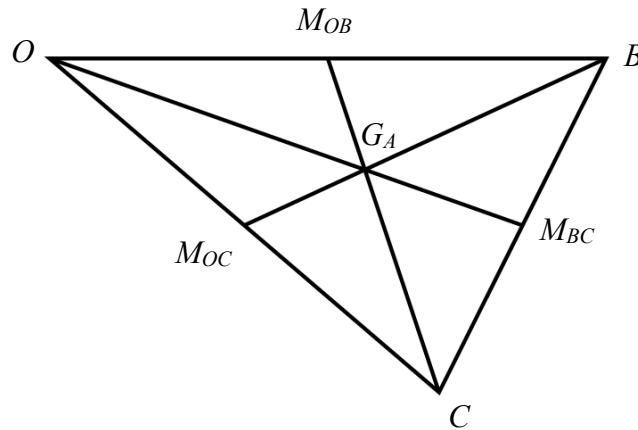
- (i) Determine that the value of x for which the object has zero acceleration is $x = 0.9L$. 2
- (ii) Show that $v^2 = \frac{162k}{x} + \frac{2k}{L-x} + u^2 - \frac{328k}{L}$. 2
- (iii) By substituting $x = \mu L$ and given that $\frac{u^2 L}{k} = 80$, deduce that the object is stationary at $\mu \approx 0.67$. 3
- (iv) Briefly describe the object's motion for $t > 0$. 1

End of Question 15

Question 16 (15 marks) Answer question in a SEPARATE writing booklet.

- (a) The lines joining the vertices and the midpoint of the opposite sides of any triangle have a common intersection point called a centroid.

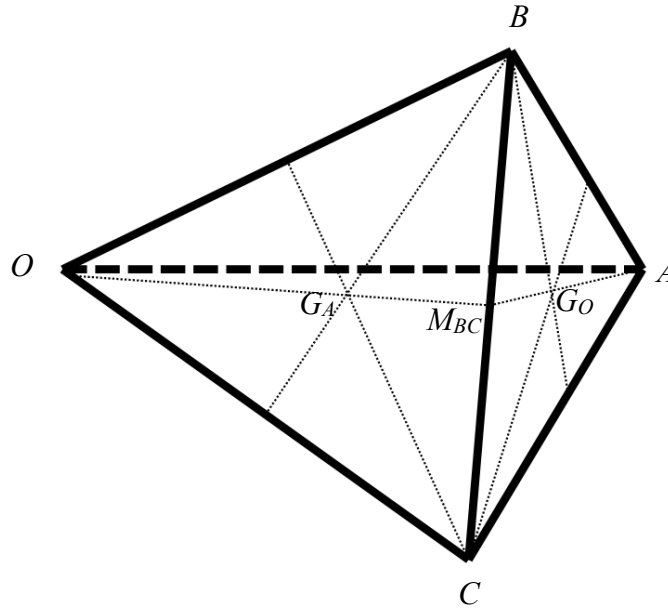
M_{OC} is the midpoint of OC , M_{OB} is the midpoint of OB and M_{BC} is the midpoint of BC .
 G_A is the centroid. $\overrightarrow{OC}$ is the vector $\underline{c}$ and $\overrightarrow{OB}$ is the vector $\underline{b}$.



- (i) Given that $\overrightarrow{OG_A} = \mu \overrightarrow{OM_{BC}}$, show that $\overrightarrow{OG_A} = \frac{\mu}{2}(\underline{b} + \underline{c})$. 1
- (ii) Show that $\overrightarrow{OG_A} = \frac{2}{3} \overrightarrow{OM_{BC}}$. 2

Question 16 continues on page 13

A triangular based pyramid has base OAC and apex B as shown. The centroid of the triangular face opposite to point A is G_A , and similarly for the other centroids. $\overrightarrow{OA} = \underline{a}$, $\overrightarrow{OB} = \underline{b}$ and $\overrightarrow{OC} = \underline{c}$. The midpoint of BC is M_{BC} .



- (iii) Show that $\overrightarrow{AG_A} = -\underline{a} + \frac{1}{3}(\underline{b} + \underline{c})$. 2

It is given that: $\overrightarrow{OG_O} = \frac{1}{3}(\underline{a} + \underline{b} + \underline{c})$ (Do NOT prove this).

- (iv) Show that the line passing through points A and G_A intersects the line passing through points O and G_O . 2
- (v) Find the position vector of the intersection point of the two lines. 1

- b) (i) Use the triangle inequality to show that 2

$$|z| < \frac{1}{2} \Rightarrow |(1+i)z^2 + iz| < 1.$$

- (ii) Is it true that 1

$$|z| < \frac{1}{2} \Leftrightarrow |(1+i)z^2 + iz| < 1? \text{ Prove your answer.}$$

Question 16 continues on page 14

- c) Suppose that there exists some natural number $n = k$, such that $n^{n+1} > (n+1)^n$.
- (i) Use a proof by contradiction to show that this must be true also for $n = k + 1$, by substituting and then multiplying the inequalities. **2**
- (ii) Find the smallest value of n for which $n^{n+1} > (n+1)^n$ holds, and then prove the inequality holds for all natural numbers greater than or equal to that value of n . **2**

End of paper

$$11. \int \frac{1}{x^2 + 6x + 13} dx = \int \frac{1}{(x+3)^2 + 4} dx$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x+3}{2} \right) + C \quad (B)$$

$$2. \theta = \cos^{-1} \left(\frac{a \cdot b}{|a||b|} \right)$$

$$= \cos^{-1} \left(\frac{-8 - 2 + 6}{\sqrt{14} \sqrt{24}} \right)$$

$$\approx 103^\circ \quad (D)$$

$$3. v = 3 + x$$

$$\frac{dv}{dt} = \frac{dx}{dt}$$

$$= v$$

$$= 3 + x$$

$$@ x = 2 \quad a = 5 \quad (C)$$



5. (D)

6. (B) (centre @ (3, 0, -2),
distance of (-3, 0, 2) from
centre is $\sqrt{6^2 + 6^2} = \sqrt{72} < 52$)

7. (C) ($a = b = 2$)

8. Roots are $\alpha, 2 \pm 3i$.

$$(2+3i)\alpha + (2-3i)\alpha + (2^2 + 3^2) = 5$$

$$4\alpha + 13 = 5$$

$$4\alpha = -8$$

$$\alpha = -2$$

$$-2 + (2+3i) + (2-3i) = p$$

$$p = 2$$

$$-2(13) = -q$$

$$q = 26$$

(D)

9. (A) (limits do not exist)

10. (C) ($\vec{u} \parallel \vec{v}$, $\vec{u} \perp$ either, $|\vec{u}| = |\vec{v}|$)

Q11 a) $z = 2 + i$, $w = 3 - 4i$

$$z\bar{w} = (2+i)(3+4i)$$

$$= 6 + 11i - 4$$

$$= 2 + 11i$$

b) $\exists n, k \in \mathbb{Z}$ such that $a = 6n$, $b = 4k$.

$$\therefore 2a + 3b = 12n + 12k$$

$$= 12(n+k)$$

$\therefore 2a + 3b$ is a multiple of 12.

c) $\int_0^{\pi/4} \tan^3 x \sec^2 x dx$

$$= \left[\frac{\tan^4 x}{4} \right]_0^{\pi/4}$$

$$= \frac{1}{4}$$

$$d) \quad I = \int e^x \sin x \, dx$$

$$u = \sin x \quad v' = e^x$$

$$u' = \cos x \quad v = e^x$$

$$I = e^x \sin x - \int e^x \cos x \, dx$$

$$u = \cos x \quad v' = e^x$$

$$u' = -\sin x \quad v = e^x$$

$$I = e^x \sin x - (e^x \cos x + \int e^x \sin x \, dx)$$

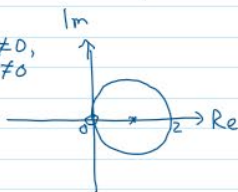
$$2I = e^x (\sin x - \cos x) + 2C$$

$$I = \frac{1}{2} e^x (\sin x - \cos x) + C$$

$$e) \quad \frac{1}{z} + \frac{1}{\bar{z}} = 1, \quad z \neq 0, \quad \bar{z} \neq 0$$

$$\frac{z + \bar{z}}{z \bar{z}} = 1$$

$$\frac{2x}{x^2 + y^2} = 1$$



$$2x = x^2 + y^2$$

$$x^2 - 2x + y^2 = 0$$

$$(x-1)^2 - 1 + y^2 = 0$$

$$(x-1)^2 + y^2 = 1$$

$$f) \quad z^3 + 8i = 0$$

$$z^3 = -8i$$

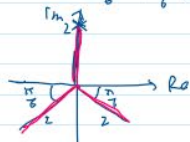
$$r^3 e^{3i\theta} = 2^3 e^{-i\frac{\pi}{2}}$$

$$r = 2, \quad 3\theta = -\frac{\pi}{2} + 2n\pi$$

$$\theta = -\frac{\pi}{6} + \frac{2n\pi}{3}$$

$$\theta = \frac{(4n-1)\pi}{6} \quad -\pi < \theta \leq \pi$$

$$\theta = -\frac{5\pi}{6}, -\frac{\pi}{6}, \frac{\pi}{6}$$



$$\text{i.e. } z = 2i, 2e^{-\frac{\pi i}{6}}, 2e^{\frac{\pi i}{6}}$$

$$12) \quad a) \quad \underline{r} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\underline{s} = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$$

Intersect if they have the same x, y, z components, i.e. if these eqns are consistent:

$$① \quad 1 + \lambda = 2 + \mu$$

$$② \quad -1 + \lambda = \mu$$

$$③ \quad 1 - \lambda = \mu^2$$

$$② + ③ \quad 0 = \mu^2 + \mu$$

$$\mu(\mu+1) = 0$$

$$\text{i.e. } \mu = 0, \lambda = 1 \text{ or } \mu = -1, \lambda = 0 \text{ (from ②)}.$$

Sub into ①:

$$\text{LHS} = 1 + 1$$

$$= 2$$

$$\text{RHS} = 2 + 0$$

$$= \text{LHS}$$

or

$$\text{LHS} = 1 + 0$$

$$\text{RHS} = 2 + -1$$

$$= \text{LHS}$$

$\therefore$ they intersect twice

b) Let $a = \frac{p}{q}$, $b = \frac{r}{t}$ where $p, q, r, t \neq 0 \in \mathbb{Z}$

Assume that $ax + b = \frac{u}{v}$ where $u, v \neq 0$
 $u, v \in \mathbb{Z}$

$$\therefore \frac{p}{q}x + \frac{r}{t} = \frac{u}{v}$$

$$ptvx + rtqv = uqt$$

$$x = \frac{uqt - rtqv}{ptv}$$

Both the numerator and denominator are integers (non-zero integers are a closed set for multiplication and subtraction). $\therefore$ RHS is rational, but x is irrational, so there is a contradiction.
 $\therefore ax + b$ is irrational.

c) $\int \frac{1}{1 + \sin \theta + \cos \theta} d\theta$ Let $t = \tan \frac{\theta}{2}$

$$\frac{dt}{d\theta} = \frac{1}{2} \sec^2 \frac{\theta}{2}$$

$$= \frac{1}{2} (1+t^2)$$

$$d\theta = \frac{2dt}{1+t^2}$$

$$\int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$= \int \frac{2dt}{1+t^2+2t+1-t^2}$$

$$= \int \frac{2dt}{2+2t}$$

$$= \ln |1 + \tan \frac{\theta}{2}| + c \quad (\theta \neq (2n+1)\pi)$$

d) RTP: $x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$

LHS - RHS: $x^{\frac{5}{2}} - x^2 - x^{\frac{1}{2}} + 1$

$$x^2(\sqrt{x}-1) - (\sqrt{x}-1)$$

$$(\sqrt{x}-1)(x^2-1)$$

Natural domain is $x \geq 0$.

$\therefore$ for $0 \leq x < 1$, $(\sqrt{x}-1) < 0$ and $(x^2-1) < 0$

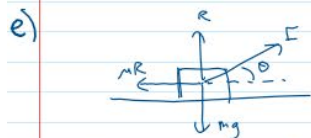
$\therefore (\sqrt{x}-1)(x^2-1) > 0$.

for $x \geq 1$, $(\sqrt{x}-1) \geq 0$ and $(x^2-1) \geq 0$.

$\therefore (\sqrt{x}-1)(x^2-1) \geq 0$

$\therefore$ LHS - RHS ≥ 0 for all real x in domain.

$\therefore x^{\frac{5}{2}} + 1 \geq x^2 + x^{\frac{1}{2}}$



Balancing horizontal forces:

$$\textcircled{1} F \cos \theta - \mu R = ma \rightarrow \mu R = F \cos \theta - ma$$

Balancing forces vertically:

$$\textcircled{2} R + F \sin \theta - mg = 0$$

From $\textcircled{1}$, $\mu = \frac{F \cos \theta - ma}{R}$

From $\textcircled{2}$, $R = mg - F \sin \theta$

$$\therefore \mu = \frac{F \cos \theta - ma}{mg - F \sin \theta}$$

13. a) For $\lambda = 1$, $S = \begin{pmatrix} 4 \\ 2 \\ 6 \end{pmatrix}$

$\therefore (4, -1, 5)$ does not lie on the line
as for $x=4$ the y & z components don't match.

b) Let $w = e^{\frac{2\pi i}{3}}$

$$1 + w + w^2 + \dots + w^{3n}$$



$$1 + w + w^2 = 0$$

$\therefore$ each sum of 3 consecutive terms is zero.

$$\begin{aligned} \therefore S &= (1 + w + w^2) + w^3(1 + w + w^2) \\ &\quad + \dots + w^{3n} \\ &= w^{3n} \\ &= 1 \end{aligned}$$

c) $\ddot{y} = -(10 + \frac{y}{10})$

$$\begin{aligned} \int_0^{y_{\max}} \frac{y}{10 + \frac{y}{10}} dy &= \int_0^{y_{\max}} -\left(10 + \frac{y}{10}\right) dy \\ \int_0^{y_{\max}} \frac{y}{10 + \frac{y}{10}} dy &= \int_0^{y_{\max}} dy \end{aligned}$$

$$y_{\max} - 0 = 10 \int_0^{y_{\max}} \frac{y}{100 + y} dy$$

$$y_{\max} = 10 \int_0^{y_{\max}} 1 - \frac{100}{100 + y} dy$$

$$= 10[y]_0^{y_{\max}} - 1000 [\ln|100 + y|]_0^{y_{\max}}$$

$$= 10y_{\max} - 1000 \ln\left(\frac{100 + y_{\max}}{100}\right)$$

$$= 10(y_{\max} - 100 \ln\left(\frac{100 + y_{\max}}{100}\right))$$

d) (i) $e^{in\theta} + e^{-in\theta} = \cos(n\theta) + i\sin(n\theta) + \cos(n\theta) - i\sin(n\theta)$
(as cosine is even & sine odd)

$$\text{OR } e^{in\theta} + e^{-in\theta} = 2\cos(n\theta)$$

$$\begin{aligned} \text{(ii)} \quad (e^{i\theta} + e^{-i\theta})^3 &= e^{3i\theta} + 3e^{i\theta} + 3e^{-i\theta} + e^{-3i\theta} \\ &= (e^{3i\theta} + e^{-3i\theta}) + 3(e^{i\theta} + e^{-i\theta}) \end{aligned}$$

$$(2\cos\theta)^3 = 2\cos 3\theta + 6\cos\theta$$

$$8\cos^3\theta = 2\cos 3\theta + 6\cos\theta$$

$$\cos 3\theta = 4\cos^3\theta - 3\cos\theta$$

$$\text{(iii)} \quad 8x^3 - 6x - 1 = 0$$

$$2(4x^3 - 3x) = 1$$

Let $x = \cos\theta$

$$2(4\cos^3\theta - 3\cos\theta) = 1$$

$$2(\cos 3\theta) = 1$$

$$\cos 3\theta = \frac{1}{2}$$

$$3\theta = \frac{\pi}{3}, \frac{5\pi}{3}, \frac{7\pi}{3}$$

$$\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$$



$\therefore x = \cos \frac{\pi}{9}, \cos \frac{5\pi}{9}, \cos \frac{7\pi}{9}$ are solutions.

(Note that the eqn must have three

(Note that the eqⁿ must have three (complex) roots. Other values of θ that satisfy $\cos 3\theta = \frac{1}{2}$, such as $\frac{11\pi}{9}$, yield the same solution for x ($\cos \frac{11\pi}{9} = \cos \frac{7\pi}{9}$))

e) $u_{n+1} = 2u_n + 3, u_1 = 1.$

Prove $u_n = 2^{n+1} - 3.$

For $n=1, u_1 = 1$

and $u_1 = 2^{1+1} - 3$
 $= 4 - 3$
 $= 1$

$\therefore$ true for $n=1$

Assume true for $n=k,$

i.e. $u_k = 2^{k+1} - 3. \quad (*)$

RTP true for $n=k+1,$

i.e. $u_{k+1} = 2^{k+2} - 3$

$u_{k+1} = 2u_k + 3$ (given)
 $= 2(2^{k+1} - 3) + 3$ (using $*$)
 $= 2^{k+2} - 6 + 3$
 $= 2^{k+2} - 3$

$\therefore$ true for $n=k+1.$

$\therefore$ true for all positive integers n
 by mathematical induction.

14. a) Contrapositive: RTP:

If n is even then $(n-5)^2$ is odd.

$\exists k \in \mathbb{Z} \mid n = 2k.$

$(2k-5)^2 = 4k^2 - 20k + 25$
 $= 2(2k^2 - 10k + 12) + 1$

$\therefore (2k-5)^2$ is odd.

$\therefore$ contrapositive statement is true

$\therefore$ if $(n-5)^2$ is even, n is odd.

b) $|\underline{a}| = 1$ by definition.

$\therefore x^2 + y^2 + \frac{7}{16} = 1$

$x^2 + y^2 = \frac{9}{16} \quad (1)$

Also, $\underline{a} \cdot \underline{b} = 0$

$x^2 - 2x + y^2 + \frac{15}{16} = 0$

$(x-1)^2 - 1 + y^2 + \frac{15}{16} = 0$

$(x-1)^2 + y^2 = \frac{1}{16}$



$\therefore x = \frac{3}{4}, y = 0$

OR $(1) - (2): 2x - \frac{15}{16} = \frac{9}{16}$

$2x = \frac{24}{16} = \frac{3}{2}$

Sub into (1): $\left(\frac{3}{4}\right)^2 + y^2 = \frac{9}{16}$

$y^2 = 0, y = 0 \therefore \underline{x = \frac{3}{4}, y = 0}$

$$d) \int \frac{x}{1+\sqrt{x}} dx \quad \text{Let } u = 1 + \sqrt{x}$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$\int \frac{(u-1)^2}{u} 2(u-1) du \quad \begin{aligned} dx &= 2\sqrt{x} du \\ dx &= 2(u-1) du \\ x &= (u-1)^2 \end{aligned}$$

$$= 2 \int \frac{(u-1)^3}{u} du$$

$$= 2 \int \frac{u^3 - 3u^2 + 3u - 1}{u} du$$

$$= 2 \int u^2 - 3u + 3 - \frac{1}{u} du$$

$$= 2 \left(\frac{u^3}{3} - \frac{3u^2}{2} + 3u + \ln(u) \right) + C$$

$$= 2 \left(\frac{(1+\sqrt{x})^3}{3} - 3(1+\sqrt{x})^2 + 6(1+\sqrt{x}) + \ln(1+\sqrt{x}) \right) + C$$

OR

$$\int \frac{x}{1+\sqrt{x}} dx \quad \text{Let } u = \sqrt{x} \quad (x \geq 0)$$

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}}$$

$$dx = 2u du$$

$$\int \frac{u^2}{1+u} 2u du$$

$$= 2 \int \frac{u^3}{1+u} du$$

$$= 2 \int \frac{u^2(u+1) - u(u+1) + (u+1) - 1}{u+1} du$$

$$= 2 \int u^2 - u + 1 - \frac{1}{u+1} du$$

$$= 2 \left(\frac{u^3}{3} - u^2 + 2u - \ln(u+1) \right) + C$$

$$= \frac{2x\sqrt{x}}{3} - x + 2\sqrt{x} - 2\ln(\sqrt{x}+1) + C$$

OR

$$\int \frac{x}{1+\sqrt{x}} dx \quad \text{Let } x = \tan^4 u$$

$$\frac{dx}{du} = 4 \tan^3 u \sec^2 u$$

$$= \int \frac{\tan^4 u}{1 + \tan^2 u} \tan^3 u \sec^2 u du \quad (\text{note } (1 + \tan^2 u) = \sec^2 u)$$

$$= 4 \int \frac{\tan^7 u}{\sec^2 u} \sec^2 u du \quad (\text{as } 1 + \tan^2 u = \sec^2 u)$$

$$= 4 \int \tan^5 u \sec^2 u - \tan^5 u du$$

$$= 4 \int \tan^5 u \sec^2 u - \tan^3 u \sec^2 u + \tan^3 u du$$

$$= 4 \int \tan^5 u \sec^2 u - \tan^3 u \sec^2 u + \tan u \sec^2 u - \tan u du$$

$$= 4 \left(\frac{\tan^6 u}{6} - \frac{\tan^4 u}{4} + \frac{\tan^2 u}{2} + \ln(\cos u) \right) + C \quad \left(\cos u = \frac{1}{\sec u} \right)$$

$$= \frac{2}{3} x^{\frac{3}{2}} - x + 2\sqrt{x} + 4 \ln \left(\frac{1}{\sqrt{1+\sqrt{x}}} \right) + C \quad = \frac{1}{\sqrt{1+\tan^2 u}}$$

$$\left(= \frac{2}{3} x^{\frac{3}{2}} - x + 2\sqrt{x} - 2 \ln(1+\sqrt{x}) + C \right)$$

$$= \frac{1}{\sqrt{1+\tan^2 u}} = (1+\sqrt{x})^{-\frac{1}{2}}$$

$$d) 400 \ddot{x} = 800 - kv^2$$

$$\begin{aligned} (i) \frac{1}{2a} \left(\frac{1}{a+b} + \frac{1}{a-b} \right) &= \frac{1}{2a} \left(\frac{a-b+a+b}{(a+b)(a-b)} \right) \\ &= \frac{1}{2a} \left(\frac{2a}{a^2-b^2} \right) \\ &= \frac{1}{a^2-b^2} \end{aligned}$$

$$(ii) v_T = 40$$

$$a = \ddot{x} = 0$$

$$0 = 800 - k(40)^2$$

$$k = \frac{800}{40^2}$$

$$k = \frac{1}{2}$$

$$400 \frac{dv}{dt} = 800 - \frac{1}{2}v^2$$

$$\int_0^v \frac{400}{800 - \frac{1}{2}v^2} dv = \int_0^t dt$$

$$800 \int_0^v \frac{1}{1600 - v^2} dv = t - 0$$

$$\frac{800}{2(40)} \int_0^v \frac{1}{40+v} + \frac{1}{40-v} dv = t$$

$$10 \left[\ln \left(\frac{40+v}{40-v} \right) \right]_0^v = t$$

$$\ln \left(\frac{40+v}{40-v} \right) = 0.1t$$

$$40+v = e^{0.1t} (40-v)$$

$$v(e^{0.1t} + 1) = 40(e^{0.1t} - 1)$$

$$v = 40 \left(\frac{e^{0.1t} - 1}{e^{0.1t} + 1} \right)$$

$$(iii) \frac{dx}{dt} = 40 \left(\frac{e^{0.1t} - 1}{e^{0.1t} + 1} \right)$$

$$\int_0^x dx = 40 \int_0^t \frac{e^{0.1t} - 1}{e^{0.1t} + 1} dt$$

$$x - 0 = 40 \int_0^t \left(1 - \frac{2}{e^{0.1t} + 1} \right) dt$$

$$x = 40 \int_0^t \left(1 + \left(\frac{2}{0.1} \right) \frac{(-0.1)e^{-0.1t}}{1 + e^{0.1t}} dt \right)$$

$$= 40 \left[t + 20 \ln(1 + e^{-0.1t}) \right]_0^t$$

$$= 40t + 800 \ln \left(\frac{1 + e^{-0.1t}}{2} \right)$$

OR

$$x = 800 \ln \left(\frac{1 + e^{0.1t}}{2} \right) - 40t$$

OR

$$x = 400 \ln \left(\frac{(e^{0.1t} + 1)^2}{4e^{0.1t}} \right)$$

OR

$$x = 400 \ln \left(\frac{(e^{0.1t} + 1)(e^{-0.1t} + 1)}{4} \right)$$

OR

$$x = 800 \ln \left(\frac{e^{0.05t} + e^{-0.05t}}{2} \right)$$

$$15.a) I_n = \int_0^{\frac{\pi}{4}} \cos^n x \, dx$$

$$(i) \text{ RTP } I_n = \frac{1}{n} \left(\frac{1}{\sqrt{2}} \right)^n + \frac{n-1}{n} \int_0^{\frac{\pi}{4}} \cos^{n-2} x \, dx$$

$$u = \cos^{n-1} x \quad v' = \cos x$$

$$u' = -(n-1) \sin x \cos^{n-2} x \quad v = \sin x$$

$$I_n = \left[\sin x \cos^{n-1} x \right]_0^{\frac{\pi}{4}} + (n-1) \int_0^{\frac{\pi}{4}} \sin^2 x \cos^{n-2} x \, dx$$

$$= \left(\frac{1}{\sqrt{2}} \right)^n + (n-1) \int_0^{\frac{\pi}{4}} (1 - \cos^2 x) \cos^{n-2} x \, dx$$

$$= \left(\frac{1}{\sqrt{2}} \right)^n + (n-1) [I_{n-2} - I_n]$$

$$I_n (1 + n - 1) = \left(\frac{1}{\sqrt{2}} \right)^n + (n-1) I_{n-2}$$

$$I_n = \frac{1}{n} \left(\frac{1}{\sqrt{2}} \right)^n + \frac{n-1}{n} I_{n-2}$$

$$(ii) \int_0^2 \frac{1}{\sqrt{(x^2+4)^3}} \, dx \quad \text{Let } \theta = \tan^{-1} \left(\frac{x}{2} \right)$$

$$x = 2 \tan \theta$$

$$\frac{dx}{d\theta} = 2 \sec^2 \theta$$

$$\text{@ } x = 2, \theta = \frac{\pi}{4}$$

$$x = 0, \theta = 0$$

$$\int_0^{\frac{\pi}{4}} \frac{2 \sec^2 \theta}{\sqrt{(4 + \tan^2 \theta)^3}} \, d\theta$$

$$= \frac{2}{4^{\frac{3}{2}}} \int_0^{\frac{\pi}{4}} \frac{\sec^2 \theta}{|\sec^3 \theta|} \, d\theta \quad \sec \theta > 0 \text{ for } 0 \leq \theta \leq \frac{\pi}{4}$$

$$= \frac{1}{64} \int_0^{\frac{\pi}{4}} \cos^3 \theta \, d\theta$$

$$(iii) \int_0^2 \frac{1}{\sqrt{(x^2+4)^3}} \, dx = \frac{I_5}{64}$$

$$I_1 = \int_0^{\frac{\pi}{4}} \cos \theta \, d\theta$$

$$= \left[\sin \theta \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{\sqrt{2}}$$

$$I_3 = \frac{1}{3} \left(\frac{1}{\sqrt{2}} \right)^3 + \frac{2}{3} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{1}{6} \left(\frac{1}{\sqrt{2}} \right) + \frac{4}{6} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{5}{6} \left(\frac{1}{\sqrt{2}} \right)$$

$$I_5 = \frac{1}{5} \left(\frac{1}{\sqrt{2}} \right)^5 + \frac{4}{5} \left(\frac{5}{6} \left(\frac{1}{\sqrt{2}} \right) \right)$$

$$= \frac{1}{20} \left(\frac{1}{\sqrt{2}} \right) + \frac{2}{3} \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{23}{60} \left(\frac{1}{\sqrt{2}} \right)$$

$$\therefore \int_0^2 \frac{1}{\sqrt{(x^2+4)^3}} \, dx = \frac{1}{64} \left(\frac{23}{60} \right) \left(\frac{1}{\sqrt{2}} \right)$$

$$= \frac{23}{3840} \left(\frac{1}{\sqrt{2}} \right)$$

b) (i) Solve $\ddot{x} = 0$:

$$0 = -\frac{81k}{x^2} + \frac{k}{(L-x)^2}$$

$$\frac{81}{x^2} = \frac{1}{(L-x)^2}$$

$$\frac{x^2}{81} = (L-x)^2$$

$$\frac{x}{9} = \pm(L-x)$$

$$x = 9L - 9x$$

$$10x = 9L$$

$$x = \frac{9}{10}L$$

$$\therefore x = \frac{9}{10}L = 0.9L$$

$$x = 9x - 9L$$

$$8x = 9L$$

$$x = \frac{9}{8}L$$

$$\text{but } 0 < x < L$$

OR

$$0 = -81(L-x)^{-2} + x^{-2}$$

$$0 = -81x^2 + 162Lx - 81L^2 + x^2$$

$$80x^2 - 162Lx + 81L^2 = 0$$

$$(10x - 9L)(8x - 9L) = 0$$

$$x = \frac{9}{10}L \text{ or } x = \frac{9}{8}L \text{ but } x < L$$

$$\therefore x = \frac{9}{10}L$$

(ii) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -81kx^{-2} + k(L-x)^{-2}$

$$\frac{1}{2} v^2 = \frac{81k}{x} + \frac{k}{L-x} + \frac{C}{2}$$

$$v^2 = \frac{162k}{x} + \frac{2k}{L-x} + C$$

@ $t=0$, $x = \frac{1}{2}L$ $v = u$:

$$u^2 = \frac{324k}{L} + \frac{4k}{L} + C$$

$$C = u^2 - \frac{328k}{L}$$

$$\therefore v^2 = \frac{162k}{x} + \frac{2k}{L-x} + u^2 - \frac{328k}{L}$$

(iii) @ $v=0$:

$$0 = \frac{162k}{mL} + \frac{2k}{L(1-\mu)} + u^2 - \frac{328k}{L}$$

$$0 = 162(1-\mu) + 2\mu + \frac{u^2 L}{k} \mu(1-\mu) - 328\mu(1-\mu)$$

$$= 162 - 162\mu + 2\mu + (80 - 328)\mu(1-\mu)$$

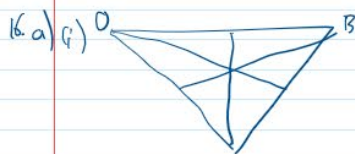
$$0 = 162 - 160\mu - 248\mu + 248\mu^2$$

$$248\mu^2 - 408\mu + 162 = 0$$

$$\mu = \frac{408 \pm \sqrt{408^2 - 4(248)(162)}}{2 \times 248}$$

$$\text{i.e. } v=0 \text{ @ } \mu \approx 0.98, 0.67$$

(iv) The object slows down until it comes to a stop @ $x \approx 0.67L$, it then changes direction and accelerates towards Earth.



$$\vec{OG}_A = \mu \vec{OM}_{BC}$$

$$\begin{aligned} &= \mu \left(\vec{OB} + \frac{1}{2} \vec{BC} \right) \\ &= \mu \left(\vec{b} + \frac{1}{2} (\vec{c} - \vec{b}) \right) \\ &= \mu \left(\vec{b} + \frac{1}{2} \vec{c} - \frac{1}{2} \vec{b} \right) \\ &= \frac{\mu}{2} (\vec{b} + \vec{c}) \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad \vec{OG}_A &= \vec{OC} + \lambda \vec{CM}_{AB} \\ &= \vec{c} + \lambda \left(-\vec{c} + \frac{1}{2} \vec{b} \right) \end{aligned}$$

Equate coeffs of (basis vectors) $\vec{b}$ & $\vec{c}$.

$$\text{b: } \frac{\mu}{2} = \frac{\lambda}{2} \Rightarrow \mu = \lambda$$

$$\text{c: } \frac{\mu}{2} = 1 - \lambda$$

Sub $\lambda = \mu$:

$$\frac{\mu}{2} = 1 - \mu$$

$$\mu = 2 - 2\mu$$

$$3\mu = 2$$

$$\therefore \mu = \frac{2}{3}$$

$$\therefore \vec{OG}_A = \frac{2}{3} \vec{OM}_{BC}$$

$$\begin{aligned} \text{(iii)} \quad \vec{AG}_A &= \vec{AO} + \vec{OG}_A \\ &= -\vec{a} + \frac{2}{3} \vec{OM}_{BC} \\ &= -\vec{a} + \frac{2}{3} \left(\vec{OB} + \frac{1}{2} \vec{BC} \right) \\ &= -\vec{a} + \frac{2}{3} \left(\vec{b} + \frac{1}{2} (\vec{c} - \vec{b}) \right) \\ &= -\vec{a} + \frac{1}{3} (\vec{b} + \vec{c}) \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad \vec{r}_A &= \vec{a} + \lambda \vec{AG}_A \\ &= \vec{a} + \lambda \left(-\vec{a} + \frac{1}{3} (\vec{b} + \vec{c}) \right) \end{aligned}$$

$$\vec{r}_O = \frac{\mu}{3} (\vec{a} + \vec{b} + \vec{c})$$

Equate coeffs of basis vectors:

$$1 - \lambda = \frac{\mu}{3}$$

$$\frac{1}{3} \lambda = \frac{1}{3} \mu$$

$$\frac{1}{3} \lambda = \frac{1}{3} \mu$$

Only two independent eqns therefore lines intersect:

$$\lambda = \mu$$

$$1 - \lambda = \frac{\lambda}{3}$$

$$1 = \frac{4\lambda}{3}$$

$$\lambda = \frac{3}{4}$$

$$\text{(v)} \quad @ \quad \lambda = \frac{3}{4},$$

$$\vec{r}_A = \vec{a} \left(1 - \frac{3}{4} \right) + \frac{1}{4} \vec{b} + \frac{1}{4} \vec{c}$$

i.e. centroid of pyramid has position vector $\vec{r} = \frac{1}{4} (\vec{a} + \vec{b} + \vec{c})$

$$\begin{aligned}
 b) (i) \quad | (1+i)z^2 + iz | &< | (1+i)z^2 | + | iz | \quad \left. \vphantom{| (1+i)z^2 | + | iz |}} \right\} \text{triangle inequality} \\
 &< |z|^2 + |z|^2 + |z| \\
 &< \left(\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2 + \frac{1}{2} \quad \text{as } |z| < \frac{1}{2} \\
 &= 1 \\
 \therefore | (1+i)z^2 + iz | &< 1
 \end{aligned}$$

(ii) Substitute $z = -\frac{1}{2}$:

$$\begin{aligned}
 \left| \frac{1}{4} - \frac{1}{4}i \right| &= \sqrt{\left(\frac{1}{4}\right)^2 + \left(\frac{1}{4}\right)^2} \\
 &= \frac{1}{4}\sqrt{2}
 \end{aligned}$$

$$< 1 \quad \text{BUT } |z| \geq \frac{1}{2}$$

$\therefore | (1+i)z^2 + iz | < 1 \not\Rightarrow |z| < \frac{1}{2}$ by counterexample

$$c) (i) \quad k^{k+1} > (k+1)^k$$

Assume not true for $n = k+1$,

$$\text{i.e.} \quad (k+1)^{k+2} \leq (k+2)^{k+1} \quad (1)$$

$$(k+1)^k < k^{k+1} \quad (2)$$

$$(1) \times (2): (k+1)^{2k+2} < k^{k+1} (k+2)^{k+1}$$

$$\begin{aligned}
 ((k+1)^2)^{k+1} &< (k^2 + 2k)^{k+1} \\
 (k^2 + 2k + 1)^{k+1} &< (k^2 + 2k)^{k+1}
 \end{aligned}$$

Not true as $k^2 + 2k + 1 > k^2 + 2k$.

$$\therefore \text{true for } n = k+1, \quad \text{i.e.} \quad (k+1)^{k+2} > (k+2)^{k+1} \quad \text{if } k^{k+1} > (k+1)^k$$

$$\begin{aligned}
 (ii) \quad \text{For } n=1, \quad 1^2 &< 2^1 \\
 n=2, \quad 2^3 &< 3^2 \\
 n=3, \quad 3^4 &> 4^3
 \end{aligned}$$

i.e. true for $n=3$.

Assume true for $n=k$,

$$\text{i.e.} \quad k^{k+1} > (k+1)^k$$

$\therefore$ true for $n=k+1$,

$$\text{i.e.} \quad (k+1)^{k+2} > (k+2)^{k+1} \quad \text{from (i)}$$

$\therefore$ true for all $n \geq 3$ by mathematical induction.